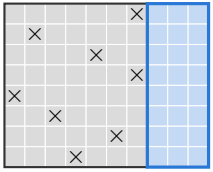
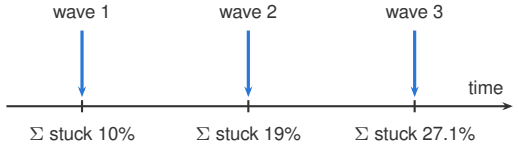
(a) Crossbar tile



active columns spares

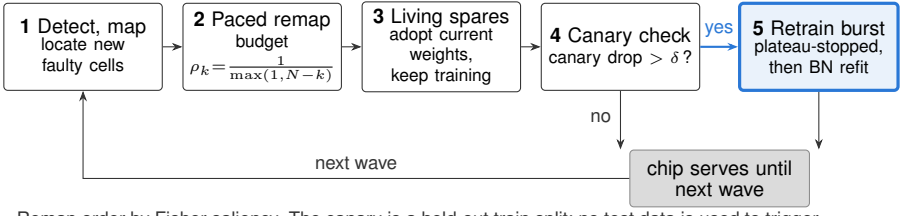
× stuck cell; 15% spare cells

(b) Fault waves over device lifetime



Each wave sticks 10% of cells. The 15% spare pool is smaller than the 27.1% cumulative total, so remapping alone cannot cover the device lifetime.

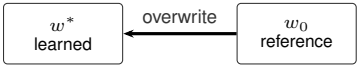
(c) Controller D, executed at every wave



Remap order by Fisher saliency. The canary is a held-out train split; no test data is used to trigger.

(d) The overwrite bug

Frozen reference (C): bug



Remap copies the reference value captured once at start, erasing what retraining learned in that cell.

Living reference (C′, D): fix

The spare takes the current weight and keeps training in later bursts; stuck weights are frozen.

Measured (old burst rate, 3 seeds):
C′ minus C = +2.48 points.